Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

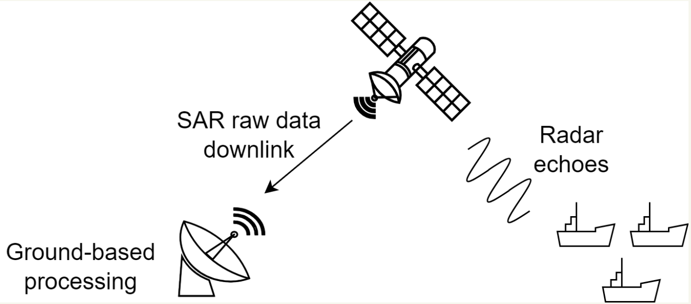
Date: 2026-08-24
Presented by: Jen-Tse Chen
Advisor: Pei-Yun Tsai

Outline

- Introduction
- Problem and Motivation
- Highlight of Contribution
- System Definition
- Previous Methods
- Proposed Method
- Comparison
- Conclusion

Introduction

- SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- Detected ships provide information for maritime surveillance.

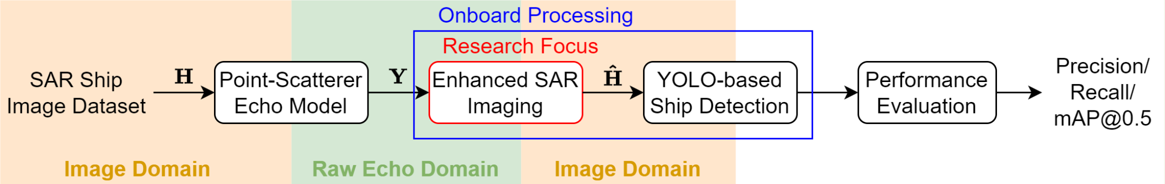


Problem and Motivation

- Ground-Based Processing
  - Large raw-data volume
  - Limited downlink bandwidth
  - Delayed target information
- Onboard Processing
  - Reduced data transmission and detection latency
  - Challenge: Limited computation, memory, and power
- Research Goal
  - Lightweight SAR processing for low-latency ship detection

Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection

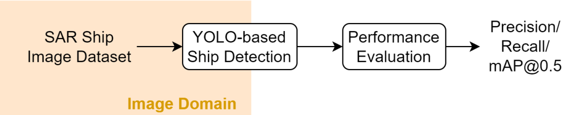


- Evaluation Metrics
  - Precision: $\frac{TP}{TP+FP}$
  - Recall: $\frac{TP}{TP+FN}$
  - mAP@0.5: Mean AP across classes at IoU threshold = 0.5

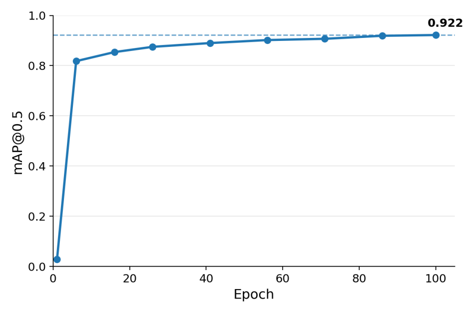
TP: True Positive, FP: False Positive, FN: False Negative

Dataset and Detection Model Setup

- HRSID
  - High-Resolution SAR Images Dataset
  - Training: 3,642 images
  - Validation: 1,962 images
  - Image size: 800 $\times$ 800 pixels
- YOLO Detection Model
  - YOLO11m fine-tuned on HRSID (100 epochs)



- Validation Performance
  - Precision: 0.910
  - Recall: 0.840
  - mAP@0.5: 0.922

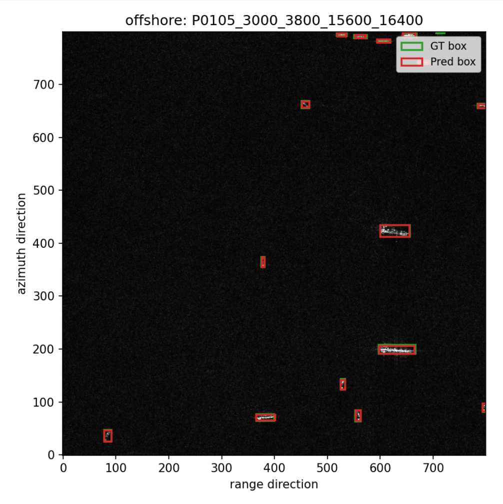


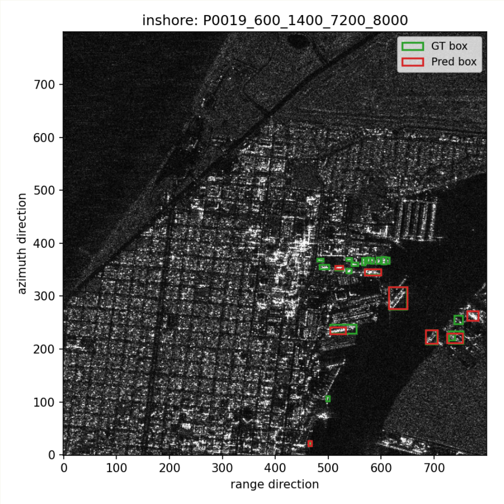
YOLO Ship Detection Results (1)

	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

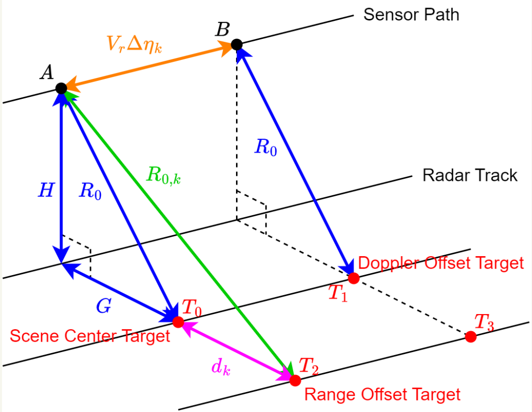
- Offshore scenes achieve better detection performance than inshore scenes.

YOLO Ship Detection Results (2)





Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta\eta_k)^2 V_r^2}$
  - $R_{0,k} \approx R_0 + d_k$
  - R_0: slant range to scene center
  - d_k: ground-range offset of scatterer k, approximated as slant-range offset
  - $\eta_m = m\Delta\eta$: slow time
  - $\Delta\eta_k$: azimuth-time offset of scatterer k

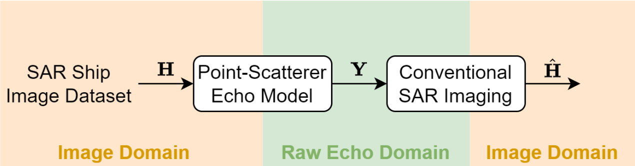
Point-Scatterer Echo Model: Signal Synthesis

- Geometry-Dependent Delay and Phase
  - $\tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}$, $\phi_k(\eta_m) = -4\pi\frac{R_k(\eta_m)}{\lambda}$
- Echo Synthesis: **S** $= [s_{m,n}]_{M\times N}$

$$s_{m,n} = \sum_k A_k \,\mathrm{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r(\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

  - $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time
  - A_k: scattering coefficient of scatterer k, $K_r = \frac{B_r}{T_r}$

Conventional SAR Imaging



- Imaging goal: Recover **H** from **Y**.
- Conventional methods: RDA, CSA, and BP

One-Step Sparse Image Refinement

- Image-domain regularization

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \underbrace{\frac{1}{2}\|\mathbf{H} - \mathbf{H}_{\text{CSA}}\|_F^2}_{\text{Preserve CSA image}} + \lambda \underbrace{\|\mathbf{H}\|_1}_{\text{Promote sparsity}}$$

  - Strong scatterers $\rightarrow$ retained; weak components $\rightarrow$ suppressed
  - $\hat{\mathbf{H}} = \mathcal{S}_\lambda(\mathbf{H}_{CSA})$, $\mathcal{S}_\lambda(x) = \frac{x}{|x|}\max(|x| - \lambda, 0), x \neq 0$
- One-Step Refinement
  - $\mathbf{Y} \xrightarrow{\text{CSA}} \mathbf{H}_{\text{CSA}} \xrightarrow{\mathcal{S}_\lambda(.)} \hat{\mathbf{H}}$

Parameters

Parameter	Value	Parameter	Value
Wavelength	0.1152 m	Closest Slant Range	4000 m
Pulse Width	12.5 μs	Height	0 m
Pulse Repetition Frequency	1 kHz	Platform Velocity	120 m/s
Bandwidth	50 MHz	Antenna Length	1.2 m
Sampling Frequency	64 MHz		

Prediction Result

Slicing

Conclusion

Reference